

segregation: Segregation Analysis, Inference, and Decomposition in Python

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Software

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Summary

segregation is an open-source Python library for measuring, analyzing, inferring, and decomposing social and spatial segregation. Developed within the PySAL ecosystem, it provides a comprehensive framework for quantifying segregation across demographic, socioeconomic, and geographic contexts using both aspatial and spatially explicit measures. The package implements over 40 single-group, multigroup, local, and spatial segregation indices, capturing dimensions such as evenness, exposure, concentration, clustering, centralization, and spatial proximity. It supports multiple spatial representations, including adjacency, distance-based, and network-based relationships.

Beyond point estimation, segregation provides Monte Carlo inference for single-value and comparative hypothesis testing under alternative null models, as well as decomposition methods that separate differences in segregation into demographic composition and spatial structure. Built for Pandas and GeoPandas, segregation integrates seamlessly with the PySAL ecosystem (S. J. Rey et al., 2022; S. J. Rey & Anselin, 2007), providing transparent, reproducible, and extensible tools for segregation research and spatial data science.

Statement of need

Residential segregation is a central topic in urban studies, sociology, demography, geography, and public policy. Although demographic and spatial data are increasingly available, existing software often provides a limited set of segregation measures, relies on proprietary platforms, or lacks support for modern inference and decomposition methods. Within Python, segregation analysis has historically depended on custom implementations despite the widespread adoption of Pandas and GeoPandas.

segregation addresses these limitations by providing:

- A comprehensive, well-documented API for segregation analysis
- Native integration with Pandas, GeoPandas, and the PySAL ecosystem
- Over 40 aspatial, spatial, local, and multigroup segregation indices
- Multiscalar segregation profiles
- Simulation-based inference for hypothesis testing
- Decomposition of segregation into demographic and spatial components
- Transparent, reproducible, and extensible implementations

These capabilities establish segregation as a unified, open-source framework for reproducible segregation analysis in Python.

State of the field

segregation is part of the PySAL ecosystem, where it provides tools for residential segregation analysis within the **Explore** family of spatial analysis libraries. Unlike desktop GIS software, it supports reproducible, scriptable workflows that integrate naturally with broader Python data science pipelines.

Although segregation analysis is also available in R packages such as *OasisR* (Tivadar, 2019), *seg* (Hong, 2011), and *segregation* (Elbers, 2021), PySAL's segregation module provides a broader and more integrated framework. It includes over 40 segregation indices, supports multiscalar and network-based analyses, Monte Carlo inference, and decomposition methods, while enabling simultaneous computation of multiple indices with minimal software dependencies. As a native Python library, it integrates seamlessly with the PySAL ecosystem and modern geospatial data science workflows.

Software design

segregation is designed with attention to flexibility, usability, and interoperability between its core functions. The library is organized into different subparts such as *singlegroup*, *multigroup*, *local*, *inference*, *decomposition*, *batch*, *network*, and *dynamics*.¹ Segregation measures are built using Python classes which can be integrated in subsequent steps, such as inference or decomposition. The module is structured toward two kinds of segregation indices: 'spatially-explicit' and 'spatially-implicit'. The former includes space as part of its original formula. The latter uses the Sean F. Reardon & O'Sullivan (2004) approach to state that any segregation index is a spatial index if you transform the data properly.

Additionally, segregation is developed with testing and documentation standards consistent with the Scientific Python ecosystem, ensuring reliability and maintainability.

Core Functionality

segregation organizes its functionality around the type of segregation analysis the user is interested in, and each subpart is explained as follows.

Single and Multigroup Indices

Single-group measures assess segregation between two different groups in a given location (i.e., one group vs. everyone else). Multigroup segregation evaluates the simultaneous separation of all groups in a population (e.g., the distribution of White, Black, Asian, and Hispanic residents) across areas.

Currently, segregation has over 40 indices available which represents, to our knowledge, the broadest range of indices available for a user in any software. Also, the user can fit many indices at once with a wrapper function in the *batch* module.

Local Indices

Unlike global indices that summarize an entire metropolitan area into a single value, local indices decompose segregation to the individual geographic unit level. Using these disaggregated measures helps identify precise spatial clusters where social isolation is most acute, uncovering micro-level dynamics that global metrics often mask. Currently, segregation has seven local indexes.

¹Cortes et al. (2020) introduced segregation but many new implementations were developed recently and the API underwent a major revision.

78 Multiscalar

79 The multiscalar profile (Sean F. Reardon et al., 2008) is a tool for measuring spatial segregation
80 dynamics—the way that a segregation index changes values as the concept of a neighborhood
81 changes, and what that tells us about macro versus micro patterns of segregation. The core
82 idea is to calculate a segregation statistic, then expand the spatial scope of a neighborhood,
83 recalculate the statistic, and repeat.

84 The package has a wrapper named `compute_multiscalar_profile` which can be used in a
85 workflow to build these profiles.

86 Simulation based Inference

87 PySAL's segregation module provides Monte Carlo inference for evaluating the statistical
88 significance of segregation indices under different null hypotheses. For single-value
89 inference, it supports resampling approaches such as bootstrap, systematic, evenness, and
90 geographic_permutation. For comparative inference, it includes methods such as bootstrap
91 and composition, which generate synthetic distributions through counterfactual estimates.
92 Because different null hypotheses test distinct assumptions, their specification is critical
93 and can lead to substantially different conclusions. Likewise, not all segregation indices are
94 appropriate for every null hypothesis, particularly in comparative analyses, making careful
95 selection of both the index and inference procedure essential.

96 Decomposition

97 The PySAL segregation module implements a decomposition framework for comparative
98 segregation analysis that partitions differences in segregation into population composition and
99 spatial structure. Building on S. Rey et al. (2021), it combines spatially explicit counterfactual
100 distributions with Shapley value decomposition to quantify each component's contribution to
101 differences in segregation across cities, time periods, and spatiotemporal contexts. Applicable
102 to multiple segregation indices, the framework provides a richer interpretation than direct
103 index comparisons by identifying whether observed differences primarily reflect demographic
104 composition or neighborhood spatial organization. This functionality is available through the
105 `DecomposeSegregation` class in the decomposition subpart.

106 Example workflow

107 Assuming a pandas or geopandas DataFrame named `data_1`, a segregation index can be
108 computed as:

```
109 from segregation.singlegroup import Dissim  
  
110 seg_index = Dissim(  
    data_1,  
    group_pop_var="group_A",  
    total_pop_var="total_population",  
)
```

109 The estimated value is available through the `statistic` attribute. Statistical significance can
110 be evaluated using Monte Carlo inference:

```
111 from segregation.inference import SingleValueTest  
  
112 result = SingleValueTest(seg_index, null_approach="bootstrap")
```

111 The package also provides wrappers for computing all single-group indices (`batch_compute_singlegroup`),
112 generating multiscalar profiles (`compute_multiscalar_profile`), comparing segregation
113 measures (`TwoValueTest`), and decomposing differences into demographic and spatial
114 components (`DecomposeSegregation`).

Research impact statement

segregation is actively used in research on urban inequality, spatial demography, and education policy. Applications include comparative analyses of racial segregation (Cortes et al., 2020), measurement of spatial information theory indices across U.S. metropolitan areas (Knaap & Rey, 2024), studies of school-neighborhood segregation (S. J. Rey et al., 2024), optimization approaches for reducing school segregation (Wei et al., 2022), comparative segregation analytics (S. Rey et al., 2021), and analyses of historical redlining legacies (S. Rey & Knaap, 2022).

The package is also used in spatial data science education, including textbooks (Knaap, 2026) and instructional materials presented at conferences such as SciPy.²

AI usage disclosure

No generative AI or LLMs were used for code development in segregation; however, they were used for grammar, spelling corrections and consistency during the writing of this paper.

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segregation is developed as part of the PySAL community, which brings together researchers and developers working on spatial analysis methods and software. The project builds on decades of research in segregation, urban, and spatial data science, and benefits from contributions across the open-source geospatial community.

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